Antenna Characterization

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1 Aim

The aim of this experiment is to characterize the RF antenna by measuring received power at different transmitter–receiver separations and angles, and to study its co-polarization and cross-polarization patterns.

2 Apparatus Required

RF signal generator, transmitting and receiving antennas, spectrum analyser, RF cables and adapters, antenna stands.

3 Theory

An antenna transmits or receives electromagnetic waves. The radiation or reception pattern describes how the signal strength varies with direction. In co-polarization measurements, the transmitting and receiving antennas have the same polarization; in cross-polarization measurements, their polarizations are perpendicular. The half-power beamwidth (HPBW) is the angular separation between the two points at which the main-lobe power falls by 3 dB from its maximum. If the E-plane and H-plane beamwidths are known in radians, an approximate directivity can be estimated using

$$D \simeq \frac{4\pi}{\theta_E \theta_H}.$$

This estimate requires clear identification of the main lobe and its two half-power points.

For a quarter-wave monopole, the approximate length is

$$L = \frac{\lambda}{4} = \frac{c}{4f},$$

where c is the speed of light and f is the operating frequency.

4 Experimental Procedure

1. The antenna was operated at the selected frequency, and received power was measured at different transmitter–receiver separations.

2. For the distance measurements used here, the preamplifier was kept ON and the attenuator was set to 20 dB.

3. The co-polarization readings were recorded by keeping the antenna polarizations aligned and varying the angle.

4. The cross-polarization readings were recorded with the antenna polarizations perpendicular to each other.

5. The measured power values were plotted against separation and angle.

5 Observations and Calculations

5.1 B. Received Power versus Distance

The received power was measured at different transmitter–receiver separations. The preamplifier was kept ON and the attenuator was set to 20 dB. The measured data were compared with the ideal free-space prediction and a fit to the experimental readings.

For ideal free-space propagation, the received power varies inversely as the square of the separation distance:

$$P_r \propto \frac{1}{R^2}. \tag{1}$$

In decibel units, this relation can be written as

$$P_r(R) = P_r(R_0) - 20 \log_{10}\left(\frac{R}{R_0}\right), \tag{2}$$

where R_0 is the reference separation and $P_r(R_0)$ is the measured power at that separation.

Figure 1 compares the measured readings, the ideal free-space prediction, and the observed linear fit.

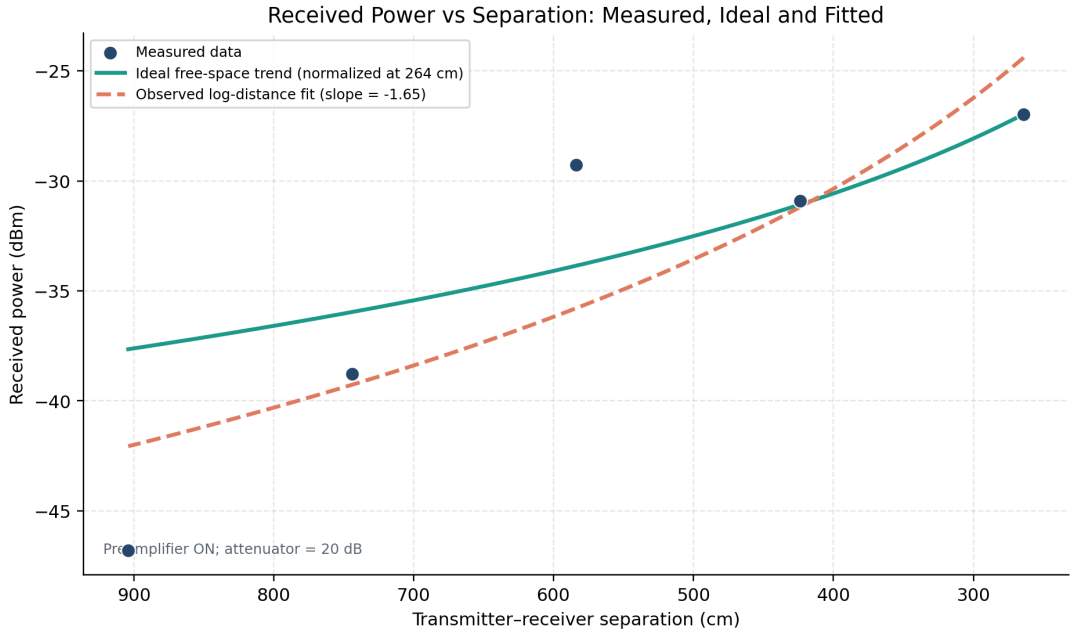


Figure 1: Measured received power, ideal free-space prediction, and observed fit as functions of transmitter–receiver separation. The preamplifier was ON and the attenuator was set to 20 dB.

The ideal curve represents the expected decrease in received power with increasing separation. The observed fit represents the trend in the experimental measurements. Differences between the measured readings and the ideal curve may arise from reflections, antenna misalignment, and measurement uncertainty.

Separation, R (cm)	Received power (dBm)
904	-46.80
744	-38.77
584	-29.27
424	-30.90
264	-26.96

Table 1: Received power measured at different separations.

5.2 C. Co-polarization Measurements

The received power was recorded at angular intervals of 10 degrees.

Table 2: Measured co-polarization power in the E-plane and H-plane.

Angle (°)	E-plane Power (dBm)	H-plane Power (dBm)
0	-27.40	-44.70
10	-27.34	-44.66
20	-27.60	-43.91
30	-28.51	-42.93
40	-29.30	-42.75
50	-30.91	-42.05
60	-32.73	-41.69
70	-34.20	-40.93
80	-36.91	-40.88
90	-41.20	-41.03
100	-41.90	-42.90
110	-40.50	-51.52
120	-41.70	-52.30
130	-39.90	-55.20
140	-39.12	-46.91
150	-41.80	-47.70
160	-46.50	-47.57
170	-46.60	-49.41
180	-46.10	-53.76
190	-52.76	-58.93
200	-41.60	-51.51
210	-41.70	-50.41
220	-40.30	-50.90
230	-40.00	-52.88
240	-40.93	-53.50
250	-42.20	-53.54
260	-39.36	-54.15
270	-38.10	-50.95
280	-36.66	-44.40
290	-34.23	-41.69
300	-32.50	-40.30
310	-31.43	-39.88
320	-30.40	-38.49
330	-30.18	-39.70
340	-28.77	-41.20
350	-28.17	-43.17
360	-27.62	-44.74

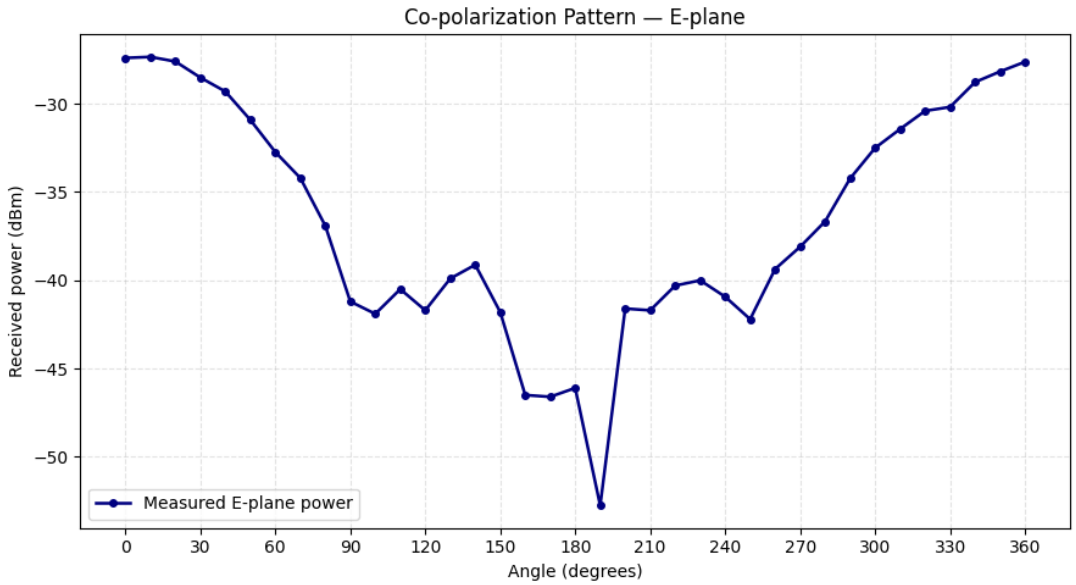


Figure 2: Measured co-polarization response in the E-plane.

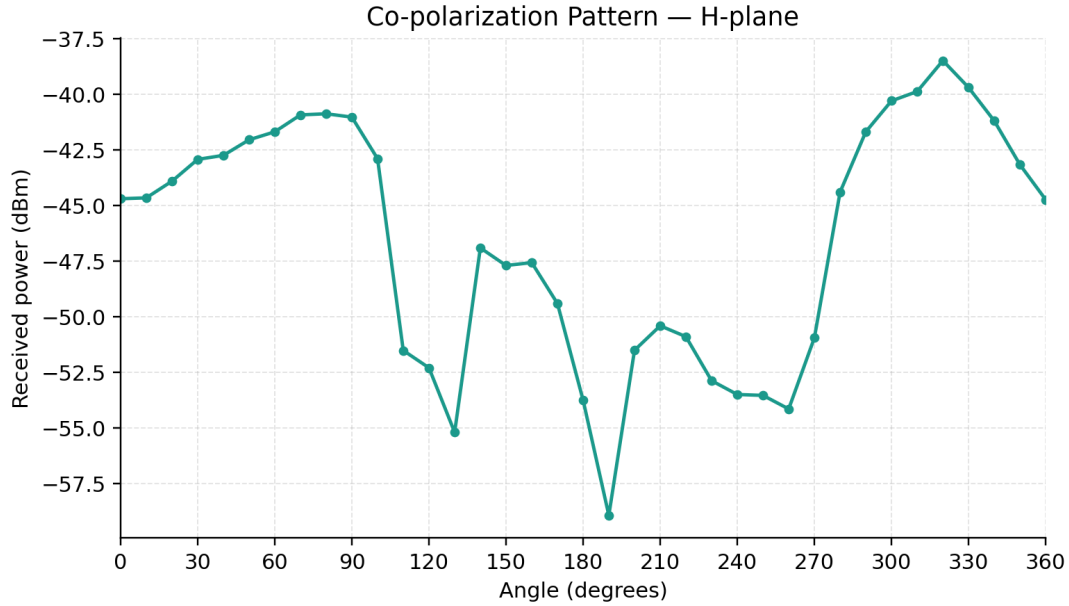


Figure 3: Measured co-polarization response in the H-plane.

5.3 D. Cross-polarization Measurements

The cross-polarization readings were recorded at intervals of 20 degrees.

Table 3: Measured cross-polarization power in the E-plane and H-plane.

Angle (°)	E-plane Power (dBm)	H-plane Power (dBm)
0	-62.68	-27.76
20	-50.50	-28.32
40	-48.06	-30.80
60	-48.62	-32.12
80	-48.80	-32.92
100	-52.88	-33.53
120	-49.37	-39.61
140	-43.10	-41.91
160	-54.50	-47.67
180	-58.50	-40.89
200	-58.62	-47.56
220	-56.67	-41.84
240	-56.40	-44.70
260	-52.70	-45.67
280	-43.11	-57.25
300	-43.00	-40.80
320	-44.15	-32.50
340	-47.81	-29.51
360	-67.13	-27.66

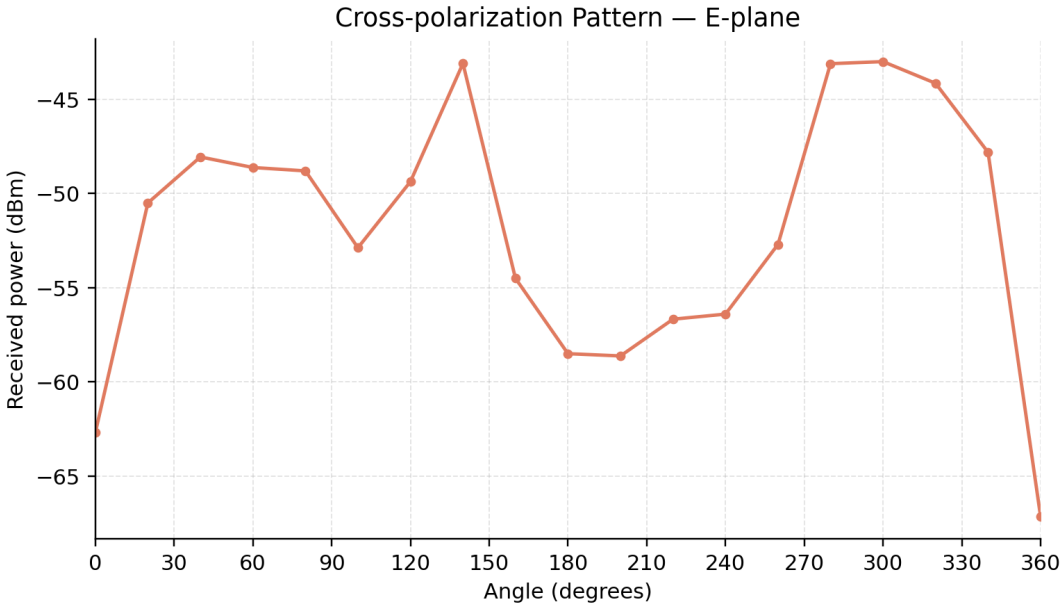


Figure 4: Measured cross-polarization response in the E-plane.

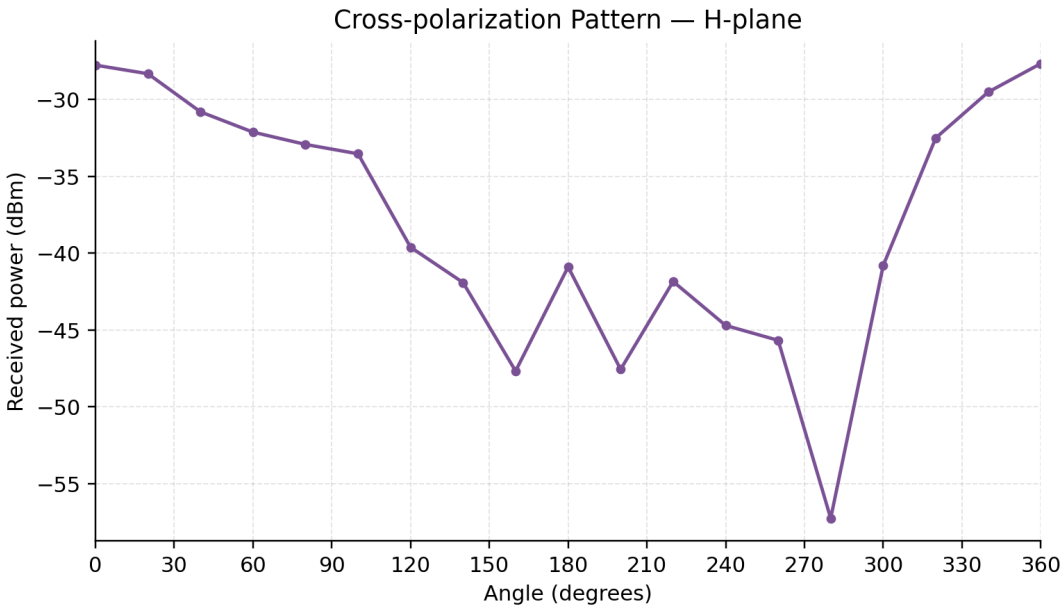


Figure 5: Measured cross-polarization response in the H-plane.

5.4 Part E: Half-Power Beamwidth and Directivity

The half-power beamwidth (HPBW) is the angular separation between the two points at which the received power falls by 3 dB below the maximum power of the main lobe.

5.4.1 E-plane HPBW

From the co-polarization E-plane measurements, the maximum received power is

$$P_{\max,E} = -27.34 \text{ dBm}. \tag{3}$$

The half-power level is

$$\begin{aligned} P_{-3\text{dB},E} &= P_{\max,E} - 3 && (4)\\ &= -27.34 - 3 && (5)\\ &= -30.34 \text{ dBm}. && (6) \end{aligned}$$

The two crossing angles are approximately $\theta_1 = 322.7^\circ$ and $\theta_2 = 46.5^\circ$. Since the main lobe crosses the $0^\circ/360^\circ$ boundary,

$$\begin{aligned} \theta_E &= (360^\circ - \theta_1) + \theta_2 && (7)\\ &= (360^\circ - 322.7^\circ) + 46.5^\circ && (8)\\ &\approx 83.8^\circ. && (9) \end{aligned}$$

Converting into radians,

$$\theta_E = 83.8\frac{\pi}{180} \approx 1.46 \text{ rad}. \tag{10}$$

5.4.2 H-plane HPBW

The maximum received power in the H-plane is

$$P_{\max,H} = -38.49 \text{ dBm}. \tag{11}$$

The corresponding half-power level is

$$\begin{aligned} P_{-3\text{dB},H} &= P_{\max,H} - 3 && (12)\\ &= -38.49 - 3 && (13)\\ &= -41.49 \text{ dBm}. && (14) \end{aligned}$$

The two crossing angles are approximately $\phi_1 = 291.4^\circ$ and $\phi_2 = 341.5^\circ$. Therefore,

$$\begin{aligned} \theta_H &= \phi_2 - \phi_1 && (15)\\ &= 341.5^\circ - 291.4^\circ && (16)\\ &\approx 50.1^\circ. && (17) \end{aligned}$$

Converting into radians,

$$\theta_H = 50.1 \frac{\pi}{180} \approx 0.874 \text{ rad}. \tag{18}$$

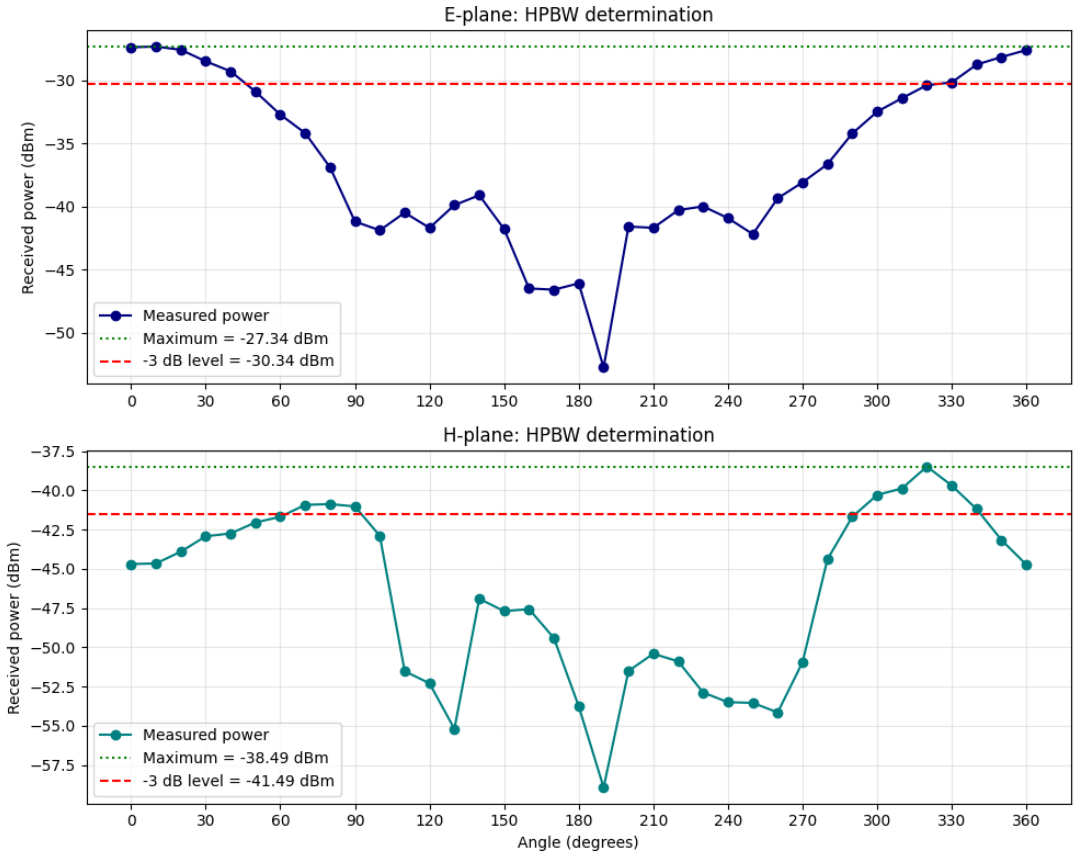


Figure 6: Co-polarization E-plane and H-plane patterns showing the measured maximum power and the corresponding -3 dB levels used for estimating the half-power beamwidth.

5.4.3 Estimation of Directivity

The approximate directivity is calculated using

$$D \approx \frac{4\pi}{\theta_E \theta_H}, \tag{19}$$

where both beamwidths are expressed in radians.
Substituting the values,

$$D \approx \frac{4\pi}{(1.46)(0.874)} \tag{20}$$

$$\approx 9.84. \tag{21}$$

The directivity in dBi is

$$D_{\text{dBi}} = 10 \log_{10}(D) \tag{22}$$

$$= 10 \log_{10}(9.84) \tag{23}$$

$$\approx 9.93 \text{ dBi}. \tag{24}$$

Table 4: Estimated HPBW and directivity.

Quantity	Estimated value
E-plane HPBW	83.8°
E-plane HPBW (radians)	1.46 rad
H-plane HPBW	50.1°
H-plane HPBW (radians)	0.874 rad
Directivity	9.84
Directivity (dBi)	9.93 dBi

The calculated directivity is an approximate estimate based on the measured beamwidths. Its accuracy is limited by irregularities in the experimental patterns and uncertainty in identifying the main-lobe half-power points.

6 Results

- Received power was measured at five transmitter–receiver separations with the preamplifier ON and attenuator set to 20 dB.
- Co-polarization responses were recorded in the E- and H-planes at 10-degree intervals.
- Cross-polarization responses were recorded in the E- and H-planes at 20-degree intervals.

- The half-power beamwidth was estimated from the measured co-polarization patterns. The E-plane HPBW was approximately 83.8°, while the H-plane HPBW was approximately 50.0°. Using these values, the estimated directivity was 9.84, corresponding to approximately 9.93 dBi. These values are approximate because the measured radiation patterns show irregularities.

7 Conclusion

The antenna was characterized by measuring received power at different separations and angles in co-polarized and cross-polarized configurations. The measured patterns show that the received power depends on antenna orientation.

8 References

Department Laboratory Manual, "Antenna Characterization," SPA617 Experimental Manual.

Statement on the Use of AI

AI tools were used as an assistive tool for language refinement, LaTeX formatting, and graph presentation. The experimental readings were taken during the laboratory experiment.